

PRADEEP GANDU

Cloud Data Engineer · Automotive Software & AI/ML

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Master's student in Automotive Software Engineering at TU Chemnitz (90/120 ECTS) with a strong engineering focus in cloud data engineering (GCP/AWS), distributed stream processing (Apache Beam, PySpark), and autonomous vehicle systems (CARLA, CAN-Bus UDS). Proven track record developing production-style streaming pipelines, explainable vehicle diagnostics audited on 39k+ real OBD records, and automated CI/CD workflows. Eligible for Master Thesis, Working Student, and Full-Time roles in Germany.

TECHNICAL SKILLS

Cloud & Big Data:	GCP (BigQuery, Pub/Sub, Dataflow, Dataproc, Cloud Storage, IAM), AWS (EC2, S3, RDS), Apache Beam, PySpark, Spark SQL, Apache Airflow, HDFS, Parquet.
Software & Tools:	Python, SQL, FastAPI, Docker, Docker Compose, Pydantic, Streamlit, Linux/Bash, Git/GitHub Actions CI/CD, Pytest, Ruff, MyPy.
Automotive & AI:	CARLA Simulator, 3D Trajectories, CAN-Bus, UDS (ISO 14229), DBC Networks, SocketCAN, IsolationForest, RAG, Ollama (Qwen3), Whisper STT, YOLOv8.

SELECTED TECHNICAL PROJECTS

Automotive Telemetry Streaming Platform (GCP)	[GCP Pub/Sub, Apache Beam / Dataflow, BigQuery, Airflow, Pytest CI]	2026 · GitHub
- Architected an event-driven streaming pipeline ingesting CARLA-style telemetry using Pub/Sub attributes and event-time semantics in Apache Beam. - Implemented stateful event_id deduplication, JSON schema validation, and automated Dead-Letter Queue (DLQ) routing for corrupted payloads. - Engineered partitioned and clustered BigQuery tables with mandatory partition filters to prevent costly full-table scans. - Orchestrated daily data quality reporting with Apache Airflow in Docker Compose with automated verification in GitHub Actions CI.		
AutoSense AI: Predictive Vehicle Diagnostics & Agent	[FastAPI, IsolationForest, RAG, Ollama LLM, MCP, React / Vite]	2026 · GitHub
- Engineered a pre-fault diagnostic system combining IsolationForest anomaly detection and personalized vehicle baselines to detect degradation before DTCs. - Audited on 39,779 real OBD records; developed a 5-of-7 temporal alert stabilization filter that reduced nuisance false alarms by 93.44%. - Integrated local TF-IDF RAG retrieval for OEM repair procedures and an autonomous agent equipped with MCP tools and deterministic safety gates.		
CAN-Bus Diagnostic & Health Simulator	[python-can, UDS (ISO 14229), DBC Network, Streamlit, SocketCAN]	2026 · GitHub
- Simulated multi-ECU CAN communication (Engine, Dynamics, Battery) using DBC definitions with modulo-16 alive counters and checksum verification. - Implemented UDS application services (0x10 Session Control, 0x22 Read Data By ID, 0x19 DTC Info) with automated YAML fault injection and Streamlit UI. - Maintained >70% branch test coverage with strict Ruff, Black, and MyPy quality gates executed in Linux/Windows GitHub Actions CI.		
Scenario-Centric AD Dataset & Trajectory Pipeline (Master Thesis)	[CARLA 0.9.x, ADAS, Python, OpenDRIVE, Parquet]	TU Chemnitz · Current
- Generating complex highway merge and lane-change scenarios in CARLA Scenario Runner, extracting synchronized 3D actor trajectories and road topology. - Building an automated ingestion pipeline structuring raw simulation sensor frames into validated, columnar Parquet schemas for ML prediction models.		

PROFESSIONAL EXPERIENCE

Working Student - AI-Assisted Software	Jan 2025 – Jul 2025
DiPP GmbH	Germany
- Developed AI-assisted routines and automated workflows for technical data processing and systematic error analysis. - Co-authored structured technical documentation and verification procedures.	
Translator & Data Specialist	Dec 2021 – Jul 2022
Ascent Staffing (Client: Clarivate Analytics)	India
Processed large-scale technical and patent datasets with strict schema validation and quality SLA controls.	
Associate Language Specialist	Oct 2020 – Dec 2021
GlobalLogic Technologies	India
Managed structured data preparation, labeling verification, and data anomaly analysis.	
Senior Process Executive	Sep 2017 – May 2019
Cognizant Technology Solutions	India
Maintained relational operational datasets, process tracking, and QA controls.	

EDUCATION

M.Sc. Automotive Software Engineering	Current
TU Chemnitz, Germany	90/120 ECTS
Thesis: Scenario-Centric Datasets & Trajectories for AD in CARLA.	
M.Tech. Embedded Systems	2017 – 2019
MLR Institute of Technology, India	First Class (CGPA 7.33)
B.Tech. Electrical & Electronics Eng.	2009 – 2013
JNTU Hyderabad, India	First Class (62.67%)

CREDENTIALS

- Google Cloud Data Engineering Program
- Python Essentials 1 & 2 (Cisco / OpenEDG)
- SQL Certified (HackerRank)

LANGUAGES

- German: Goethe-Zertifikat B1 Certified
- English: IELTS B2 (Professional Working)
- Telugu: Native / First Language